



Computational insights into deep eutectic solvent design: Modeling interactions and thermodynamic feasibility using choline chloride & glycerol

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ABSTRACT

To address climate change and environmental concerns, the chemical process industries are increasingly adopting green technology and exploring eco-friendly waste management practices. Deep Eutectic Solvents (DES) have emerged as promising green solvents with diverse applications in the industries in separation processes, catalysis, drug design, and a lot more. However, the ideal way for modeling ionic interactions within the hydrogen-bond acceptor (HBA) during DES synthesis remains uncertain, and theoretical insight on the formation mechanism of energetically stable DESs from choline chloride (CHL) and urea (URE) is yet to be explored. In this study, we investigate different modeling approaches (no bond, generic bond, or single bond) for ionic interactions in hydrogen-bond acceptor (HBA) molecules during DES synthesis and the thermodynamic feasibility of forming an energetically stable DES using choline chloride (CHL) and urea (URE) through theoretical calculations (PM3, HF, and DFT). Our findings indicate that the choice of modeling approach does not significantly affect the outcomes across calculation levels. The H-X DES formation pathway emerges as the most viable route for CHL: GLC DES synthesis, offering strong binding energy and high exothermicity. This pathway provides a robust solution for synthesizing energetically stable DES with potential applications in various industries. Researchers can consider alternative calculation levels when DFT becomes cost-prohibitive to optimize computational costs. Our study contributes valuable insights into DES design and synthesis, promoting environmentally sustainable and cost-effective processes.

Introduction

As the promotion of green technology continues to gain significant attention with the rising climatic change and global warming, the effort towards the improvement of pollution control and the promotion of environmental friendliness of the different operations in the oil and gas sector is likewise advancing [11,64]. Several areas promoting green technology and environmental friendliness in our respective communities are primarily explored in areas of waste management involving the transformation of wastes like agro-wastes, polymeric, and many other kinds of wastes into valuable products like fuels, materials, energy, and many other applications [16,17,38,47,48,57]. Recent studies are exploring the potential of some green solvents as potential solvents that

could facilitate the extraction of certain materials of interest from a mixture. The study of the green solvents' design and synthesis techniques is significantly gaining better attention across different applications in process industries. Some of these processes include lactic acid production [43], gas absorption [24,34,58,65,68], sorption of some heavy metals and dyes [7,22,52,59], petroleum processing, enhanced oil recovery process [20,39], catalysis of chemical processes [61], recycling of wastes lithium batteries [15,33], and a lot more. Examples of such kinds of solvents include the deep eutectic solvent (DES) and ionic liquid (IL) [9,31,35]. However, this report focuses on DES exploration. For the purpose of this report, we focus on the exploration of DES's potential for varying applications. Some important features of deep eutectic solvent (DES) include less toxicity, high degradability, good biocompatibility,

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non-volatile, non-flammable, good renewability, and many other characteristics [10]. The application of deep eutectic solvent (DES) is not limited to only separation process but also to catalysis, extraction, electrolytic process, and other processes, which could be applicable to a wide range of process industries like the food, pharmaceutical, cosmetics, oil, and gas, and a lot more [5,27,32,60,66].

Some reports have categorized both of the solvents as a green solvent, while other references like Chen and Mu [9] revisited the subject of the greenness of the solvent. Their report [9] has indicated that more and more evidence in the literature is showing the non-greenness of the solvents due to some characteristics exhibited by some of them, which include instability, volatility, toxicity, flammability, and difficulty in regenerating them for further use. Chen and Mu [9] further suggest the functionalization and modification of the solvent properties to reform their behavior to improve their greenness. Many works in the literature [5,10,27,32,34,59,60,66] have been exploring the suitability of the different forms of DES applications in the different processes. Some of which includes searches focusing on the identification of the stable DES for a potential application. In such a study, a wide range of studies do focus on the screening of hydrogen-bond acceptor (i.e., HBA) and hydrogen-bond donor (i.e., HBD) that would feasibly design a solvent with a defined characteristic of interest for whatever applications. Examples of some of the HBD components include urea, glycerol, ethylene glycol, glucose, thiourea, acetamide, malonic acid, phenol, citric acid, maleic acid, succinic acid, and a lot more; and the HBA entails choline chloride, ethylammonium chloride, methyltriphenylphosphonium bromide, betaine, 1,3-butanediol, acetylcholine chloride, choline tetrafluoroborate, triethylmethylammonium chloride, levulinic acid, choline proline and many others, which are used in the formation DES [2,44].

Some of the existing studies have synthesized DES for the sorption of certain pollutants like heavy metals, VOC, and dye [7,22,27,35,52]; the catalysis of some chemical processes [61]; and the promotion of separation process efficiency. Many of these studies reported in the literature have primarily deployed experimental study approaches, while only a few studies have employed a theoretical approach to investigate its synthesis process using a computational approach. Among the few theoretical studies is the work of Naseem et al. [41] who studied the relationship between computed IR spectra for choline chloride-based deep eutectic solvents with ones experimentally collected. From which, the study confirms a good agreement with the experiment one. Ghenaarian et al. [21] studies further established that choline chloride: urea better interacts with palladium nanoparticles than the choline chloride (CHL): ethylene glycol (EGL) deep eutectic solvent (DES). Using a theoretical approach, the use of guanidine hydrochloride (GuHCl): lactic acid (LA) in the ratio of 1:1 has been established to have formed DES that could dissolve hemicellulose due to the higher interaction deduced for it from their studies of lactic acid-based DES [67]. Doherty and Acevedo [14] developed a reliable non-polarizable force field (OPLS-DES) for effective prediction of DES' density, heat capacity, surface tension, and viscosity, which shows a good agreement with the experiment. Nicolas et al. [19] also explores the role of DES in enhancing the solubility of desired component present in a mixture during extraction processes. The thermophysical and transport properties were also investigated by Cea-Klapp et al. [8] for the DESs like CHL:EGL and CHL:GLC in presence of methanol using molecular dynamics, while some work [1] investigated the impact of water and HBA on the density and viscosity of a DES for providing insight on the design and re-design of new solvent. Other works like Santra et al. [53] were theoretically able to have a stable geometry of the choline chloride (CHL): urea (URE) deep eutectic solvent (DES) as a measure for understanding the stability of DES formed from the use of HBD (URE) and HBA (CHL).

The existing literature review has not yet provided a definitive approach for modeling the ionic interaction within the hydrogen-bond acceptor (HBA) building. It remains unclear whether this interaction should be represented with no bonds, generic bonds, or single bonds. It is imperative to clearly establish whether any of these modeling

approaches can influence the outcomes of a study.

Furthermore, the computational investigation of the mechanism involved in synthesizing a highly stable choline chloride (CHL): glycerol (GL) deep eutectic solvent (DES) has not been comprehensively explored in the current body of literature. DESs created using polyol-based hydrogen bond donors (HBDs) have gained widespread industrial applications, particularly those composed of glycerol and ethylene glycol, owing to their lower freezing points and ability to remain in liquid form below room temperature [12,18,26].

To address the existing research limitations stated earlier in our survey of the literature, we, therefore, explore the different possible approaches to modeling the electrovalent or ionic interaction existing in the hydrogen-bond acceptor (HBA) molecule used in the synthesis of deep eutectic solvent (DES) to understand the agreement and the implication of using either of the approaches (using no bond, generic bond, or single bond) in a study. Furthermore, we investigate the mechanism involved with forming an energetically stable DES from the use of choline chloride (CHL) and glycerol (GLC). In addition, we later investigated the prediction agreement that existed at different levels of calculations like PM3, Hartree Fock (HF), and density functional theory (DFT). Findings made from this study serve as an established strategy for building different stable DES(s) for screening out solvents that best suit a defined application or use to be investigated at a low and high level of computational cost.

Methodology

In the study, we employed a computational approach via a quantum mechanics calculation. The later section of this report presents comprehensive details on the strategy and computational information deployed in our study.

Study strategy

In this section, we present the strategies employed in the evaluation of our chosen computational approach strengths and limitations, the assessment of the approaches used in the modeling of ionic bonds, and one used for the identification of feasible pathways leading to the formation of the deep eutectic solvent and its corresponding formation energy.

Evaluation of the des models' components used in our study

We employed an infrared spectroscopy plot computationally generated for the hydrogen bond acceptor and hydrogen bond donor to compare with one experimentally obtained from a wet laboratory. We evaluated the similarities and differences visible on their respective plot or spectra. The computationally generated spectra or plot is obtained for different levels of computations and compared with experimentally generated ones to understand the strength and limitations of the different levels of computations evaluated.

Evaluation of different ionic bond modeling techniques in HBA

With the use of semi-empirical, Hartree Fock, and density functional theory calculations, different approaches to modeling the ionic bonds present in HBA were evaluated. The approaches, which include modeling the ionic interaction without using no bond (that is, space), with the use of single bond, and generic bonds (that is, a type of bond which does not physically distinct bond formation or breaking), were all evaluated as shown in Fig. 1.

All the sets of approaches were evaluated at the different levels of computation (semi-empirical, Hartree Fock, and density functional theory calculations [51]), assessing their electronic energies and ionic bond length consistency across the three approaches evaluated. Choline chloride was taken as our HBA component for our deep eutectic solvent.

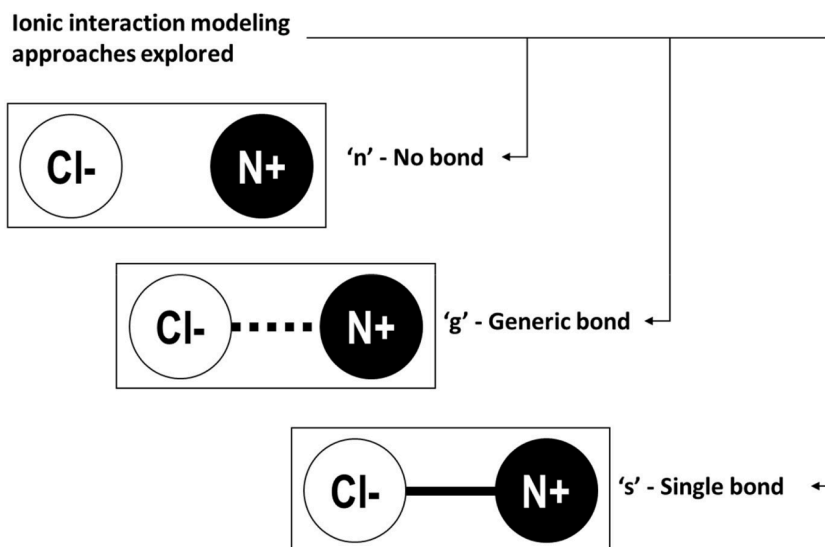


Fig. 1. A scheme showing the different ionic interaction models evaluated.

Analysis of HBD's C-C bond consistency

Further analysis of the hydrogen bond donors to understand the consistency of their bond lengths irrespective of their levels of calculation was evaluated to understand the suitability or implications of employing the use of different levels of computation for this kind of application. The semi-empirical, Hartree Fock, and density functional theory calculations were used in the analysis of glycerol as our deep eutectic solvent hydrogen bond donor.

Evaluation of the DES formation mechanism

The study evaluated the different deep eutectic solvent formation mechanisms by analyzing different interaction points on HBA (Cl, H, O, and N) with HBD (O, H, and C) with their corresponding formation reaction energy, also known as binding energy. The energies are computed using the expression in Eq. (1).

$$BE = E_{des} - E_{HBA} - E_{HBD} \quad (1)$$

The E_{des} is the electronic energies of the DES, E_{HBA} is the electronic energies of the hydrogen bond acceptor, and E_{HBD} is the electronic energies of the hydrogen donor. All the electronic energies are collected in eV. The energy that is most exothermic, shown with the highest negative value, would signify the best and most feasible interaction points.

In our study, we calculated both uncorrected and BSSE-corrected binding energies to comprehensively assess their differences and errors. We have chosen to utilize the corrected binding energies (referred to as BE_{corr}) for our subsequent analysis concerning the formation route stability of Deep Eutectic Solvents (DES). Detailed information regarding the BSSE correction methodology employed in our energy calculations to derive BE_{corr} is provided in Section 2.2 under 'Computational Details'.

Thermodynamic feasibility of the different DES formation pathways

The various thermodynamic parameters such as enthalpy, entropy, and Gibbs free energy were all computed use the generic mathematical expression in Eq. (2).

$$dX = X_{des} - X_{HBA} - X_{HBD} \quad (2)$$

The subject dX denotes the formation X , where the X could be either H, S or G like dH , dS or dG . X_{des} , X_{HBA} , and X_{HBD} terms represent the X values for the DES formed, HBA, and HBD compound. The enthalpy (H), entropy (S), and Gibbs free energy (G) are all computed standard temperature (298 K) and pressure (1 atm).

Computational details

Using the Spartan v20 molecular modeling package, we employed three different levels of computations in our study. The levels including semi-empirical calculations, which we utilized to compute infrared spectra, electronic energies, and other relevant properties used in the methods comparison was PM3, Hartree Fock calculation was also employed using 6-31G* basis set in the calculations of infrared spectra, electronic energies, and other properties. Another level of calculations we employed in our study is density functional theory calculations using B3LYP hybrid functional and 6-31 G in calculating infrared spectra, electronic energies, and other relevant parameters, in accordance with literature [25]. A Dell Precision 3520 mobile workstation with RAM of 24 GB, a processor capacity of 7 cores, and a storage of 1 TB SSD was employed to run our calculations.

To improve on the accuracy of the energies obtained from the geometry optimization calculations by correcting the basis set superposition error (BSSE), we employed the use of single-point energy calculation using DFT-B3LYP hybrid function and dual basis set [cc-pVTZ [6-31G*], via a combination a correlated corrected basis set [cc-pVTZ] and pople double-eta basis set [6-31G*]) to re-evaluate our binding energies in line with the literature [29,63], which suggests the use of dual basis set as one of measure for correcting the BSSE in our package. The outcome of employing this approach was designated as the BSSE-corrected binding energy (BE_{corr}), whereas binding energies obtained without its application were labeled as uncorrected binding energies (BE_{uncorr}).

Results and discussions

Here we present the results obtained from our study for the analysis of the hydrogen bond donor (HBD) and the hydrogen bond acceptor (HBA) involved in the formation of CHL:GLC deep eutectic solvent. The validation of the model, analysis of possible interaction mechanisms promoting the formation of DES, and many other results are presented therein.

DES model components' analysis

In our study, we further investigate the accuracy of our model via the comparative analysis of the infrared (IR) spectra obtained for the HBA and HBD employed in the formation of our DES using different levels of computation (PM3, HF, and DFT). The computed glycerol (GCL) and

choline chloride (CHL) spectra are presented in Figs. 2 and 3. Findings made from the analysis were employed to understand the accuracy of the different level of calculation used for the IR spectra computation in our study.

The analysis of the CHL spectra in Fig. 2 reveals that the spectra generally present the peaks found across the different levels of computations (PM3, HF, and DFT) and show significant peaks around wavelengths of 2700–3700 and 700–2000 1/cm. However, the wavelength range of 2700–3700 1/cm in the IR spectra computed in Fig. 2 shows that the HF gives a better agreement with DFT peaks, while PM3 shows better agreement or relationship with DFT peaks in the 700–2000 1/cm range. Collection of the significant IR spectra peaks obtainable from the PM3, HF, and DFT calculations in relation to the spectra obtained from an experimental study [3] are presented in Table 1.

Further evaluation of the peaks collected for choline chloride (CHL) from the respective calculations in Table 1 reveals the presence of different forms of functions and vibrations like C–H bending, which was obtained in PM3 at 807 and 903 1/cm, HF at 932 1/cm, and DFT at 871 1/cm, following the literature [36] that state C–N to have existed within the range of 755–880 1/cm. Other functional vibrations like C–O stretch were obtained in PM3 at 1016 1/cm, HF at 1025 1/cm, and DFT at 1004 1/cm; C–N stretching was observed in PM3 at 1093, 1139 1/cm, HF at 1032 1/cm, and DFT at 1113 1/cm; O–H bending observed in PM3 at 1376 1/cm, HF at 1433 1/cm, DFT at 1455 1/cm. Many other peaks found in the respective IR spectra in theoretically computed, and the ones experimentally collected are summarized in Table 1. Moreover, the correlation analysis of the peak values in Table 1 reveals a good agreement between theoretical calculations and the experimentally collected ones [3].

The computed infrared spectra for glycerol (GLC) molecules shown in Fig. 3 were further analyzed. The outcome of the spectra analysis reveals the presence of different peaks across the spectra shown in Fig. 3. The significant peaks identified via the analysis include the O–H stretching at 3877, 3846, and 3676 1/cm; C–O stretching of the primary alcohol at 1070, 1017 & 1167, and 1099 1/cm; O–H stretching and bending at 1410, 1531, and 1445 1/cm; and CH₂ stretching at 2905, 3297–3212, and 3002–3100 1/cm for PM3, HF, and DFT respectively, in accordance with the literature [36,28,56].

Table 2 summarizes a list of significant peaks identified from the theoretical calculation approaches and the ones experimentally collected. The spectra patterns shown in Fig. 3 and the respective peaks presented in Table 2 for the glycerol molecule's PM3, HF, and DFT calculations were generally found to have shown similar trend with one

reported for the experiment as reported in Table 2 for all the levels of the calculations, exception of few peaks that shows different wavelength (650–800 1/cm and 3600 1/cm from experimentally reported spectra [42] but still within the range of wavelength the functional groups for the O–H stretching (3548–3700 1/cm) and C–H bending (755–880 1/cm) [36,28,56].

Generally, spectra analysis to validate our chosen model's accuracy has demonstrated a good relation with those obtained from experimental studies [3,42]. Using the results of their correlation analysis with the experiment, a trend of their agreement or average correlation coefficient was found to be in the order DFT (0.9991) > HF (0.9980) > PM3 (0.9945). The analysis further reveals the DFT method agrees more with the experimental FTIR [3,42] with little disparity, although more expensive than other methods assessed. The findings made for the experimentally collected spectra with the computed spectra confirming a good agreement was similar with what was obtained for the use of similar approach in the report of Oyegoke et al. [45]. An appreciable relationship was obtained for all the computational methods employed in our analysis of the CHL and GLC, where the potential of PM3 and HF approach to complement the use of DFT calculations or management of computational time in the analysis of large structure (that could be expensive to compute) was found to be significantly good and feasible due to the relation of its peaks with ones obtained from the laboratory studies shown in their correlation coefficients (PM3 = 0.9945, HF = 0.9980) obtained for the methods.

Analysis of HBA & its ionic bond modeling strategies

In the analysis of CHL as the HBA component of the DES primarily investigated in our study, our survey of the works in the literature has not been able to clearly present a defined insight on how to model best the ionic bond present in the HBA component of a potential DES. In our study, we first investigated the correlation of different approaches to modeling the electrovalent or ionic interaction present in the HBA. The approaches include cases where the ionic interactions in the HBA were modeled without a bond, with a single bond, and a generic bond. In this context, the generic bond signifies an undefined kind of bond, which could be either existing or un-existing. The different bonds were denoted as 'a' for without bond, 'b' for with a single bond, and 'c' for the generic bond, as presented in Tables 3–5. The correlation of the different approaches was explored using PM3 semi-empirical, Hartree Fock (HF), and density functional theory (DFT) calculations.

Analysis of the results collected from the study shows an insignificant

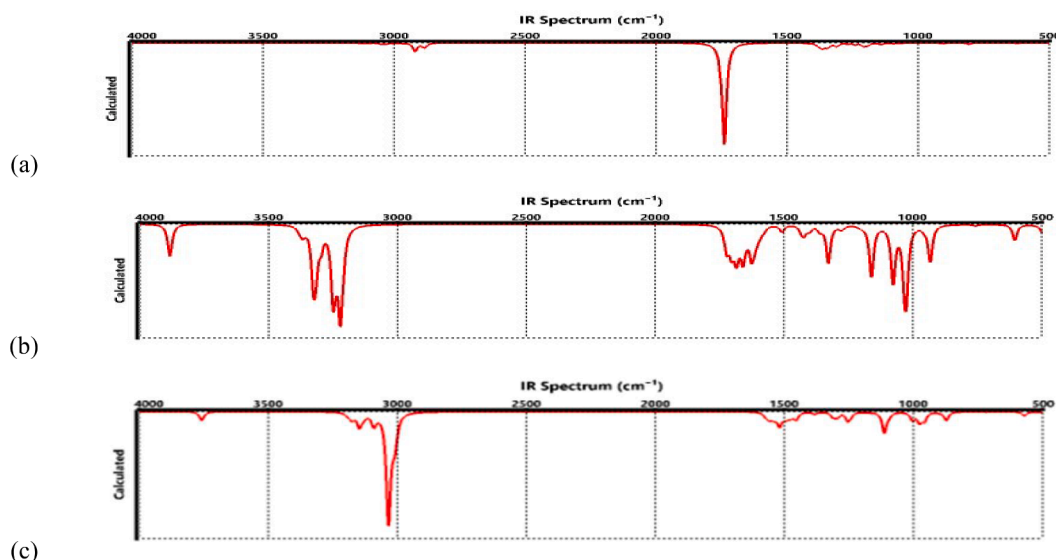


Fig. 2. IR analysis of Choline chloride using (a) PM3, (b) HF, and (c) DFT calculations.

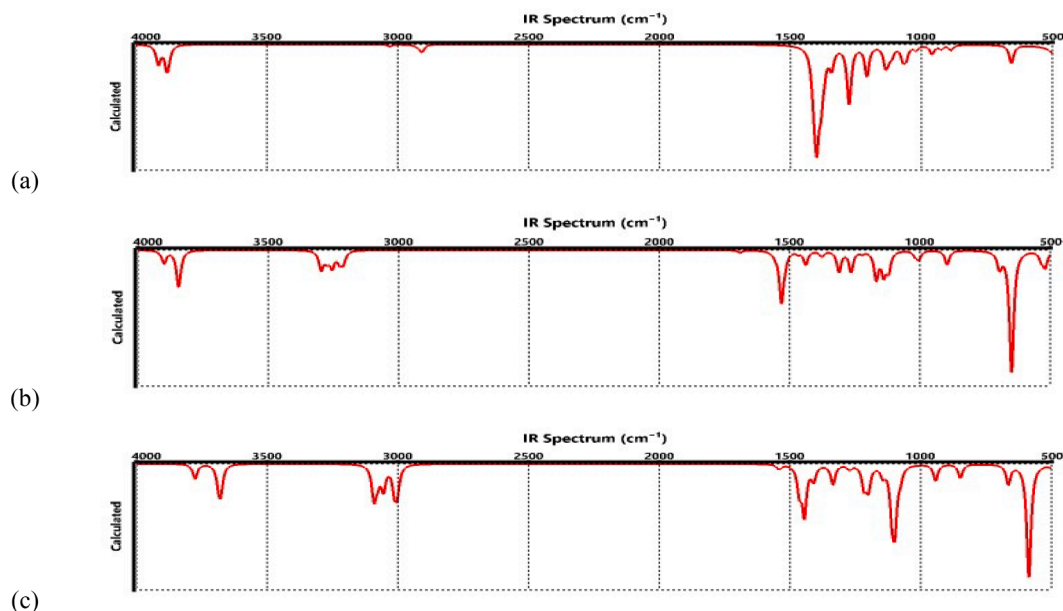


Fig. 3. IR analysis of glycerol using (a) PM3, (b) HF, and (c) DFT calculation.

Table 1

Comparison of theoretical calculation with experimental peaks for choline chloride.

Peaks [36]	PM3 (1/cm)	HF (1/cm)	DFT (1/cm)	Expt. [3] (1/cm)
CH bending (755–880)	807	–	–	807
CH bending (755–880)	903	932	871	897
C–O stretch (1000–1300)	1016	1025	1004	1000
C–N stretching (1020–1250)-medium	1093	1032	1113	1026
C–N stretching strong (1266–1342)	1139			
O–H bending (1395–1440)	1376	1433	1455	1423
Not available	1740	1616	1572	1582
CH stretching (3000–3100)	3039	3211	3032	3012
	2919			
OH stretching broad (3200–2700)	–	3315	3175	3258
OH stretching broad (3300–2500)	–	3381	–	3435
Correlation coefficient (X/Expt), where X is PM3, HF, and DFT	0.9966	0.9980	0.9989	1
	0.9943	0.9980	0.9989	1

Table 2

Comparison of theoretical calculation with experimental peak for glycerol.

Peaks ("Infrared spectroscopy correlation table - Wikipedia," n.d.; [36])	PM3 (1/cm)	HF (1/cm)	DFT (1/cm)	Expt [42] (1/cm)
C–H bending (755–880)	887–959	895	848	650–800
C–O Stretching of primary alcohol (1050–1085)	1070	1017	1099	1050
		1167		
O–H bending (1395–1440)	1410	1531	1445	1400
				1440
C–H stretching (2840–3000)	2905	3297	3002–3100	2950
		3212		3000
OH-stretching sharp (3584–3700)	3877	3846	3676	3600
Correlation coefficient (X/Expt), where X is PM3, HF, and DFT	0.9938	0.9961	0.9989	1
	0.9935	0.9999	0.9998	1

Table 3

PM3 calculation of electronic energy & bond length (n =no, g =generic, and s =single bond).

Label	HBA Formula	E (kJ/mol)	Cl–N bond length (Å)
PM3-n	Cl.C ₆ H ₁₄ NO	–245.74	3.26292662
PM3-g	C ₆ H ₁₄ ClNO	–245.74	3.26302198
PM3-s	C ₆ H ₁₄ ClNO	–245.74	3.26306649

Table 4

HF calculation of electronic energy & bond length (n =no, g =generic, and s =single bond).

Label	HBA Formula	E (eV)	Cl–N bond length (Å)
HF-n	Cl.C ₆ H ₁₄ NO	–21,289.63	3.61036140
HF-g	C ₆ H ₁₄ ClNO	–21,289.63	3.61041153
HF-s	C ₆ H ₁₄ ClNO	–21,289.63	3.61020403

Table 5

DFT calculation of electronic energy & bond length (n =no, g =generic, and s =single bond).

Label	HBA Formula	E (eV)	Cl–N bond length (Å)
DFT-n	Cl.C ₆ H ₁₄ NO	–21,472.47	3.55827636
DFT-g	C ₆ H ₁₄ ClNO	–21,472.47	3.55827622
DFT-s	C ₆ H ₁₄ ClNO	–21,472.47	3.55827630

difference in the electronic energies irrespective of the computational method employed, as shown in Table 3 for the PM3, Table 4 for the HF, and Table 5 for the DFT calculation indicating electronic energies as –245.74, –21,289.63, and –21,472.47 eV. The respective electronic energies obtained for the different computational methods have similar values except for the PM3, which has a much different value. This was due to the higher level of approximations present in the semi-empirical calculations, in accordance with the literature [51,54].

Despite the difference in the electronic energy values, a consistent trend of having the same energy across the different approaches of modeling ionic interaction (as shown in Fig. 4) in the HBA components was observed at different levels of computations carried out. Fig. 5 presents the structure of the DFT-optimized CHL obtained using

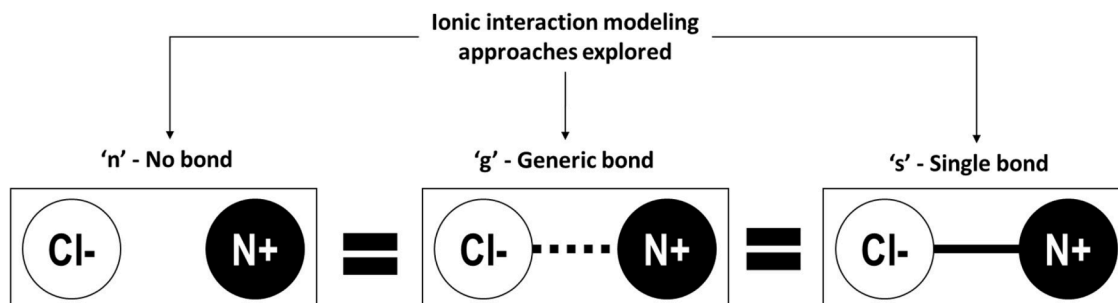


Fig. 4. A scheme showing the consistency obtained for the electronic energy and Cl-N bond length across different ionic interaction models at the different levels of calculations.

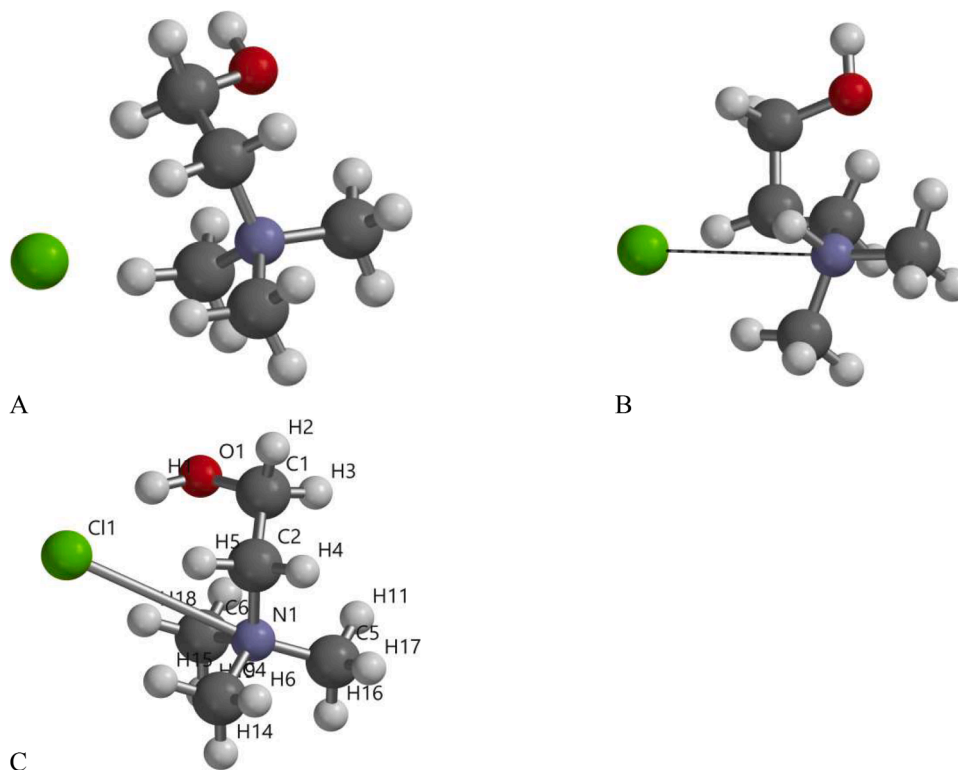


Fig. 5. The optimized geometry of choline chloride (CHL) was obtained from DFT calculation with (A) no bond, (B) generic bond and (C) single bond for the ionic interaction.

different modeling approaches for ionic interactions.

The bond length or atom distance for the anion and cation was found to be relatively close to each, irrespective of the level of the computational method employed. Table 3,4,5 indicates that PM3, HF, and DFT calculation predicts an average distance of 3.26, 3.61, and 3.56 Å for the chlorine ion (Cl-) bond or distance with nitrogen ion (N+), showing an approximate value of 3.6 Å for HF/DFT, while 3.3 Å for the use of PM3 with a slight difference 0.3 Å (see Fig. 3). The trend of showing better agreement of the DFT prediction with the HF when compared to the PM3 agreed with the literature [25].

Analysis of HBD's C-C bonds' consistencies

Here, we analyzed the glycerol (GLC) as the hydrogen bond donor (HBD) for our potential DES to be formed. The analysis was done using different computational methods like PM3, HF, and DFT. The consistency of the electronic energies and bond length existing between the different carbon-carbon bonds, including C1-C2 and C2-C3, is presented in Table 6.

Table 6

Analysis of glycerol as the HBD for the DES.

Label	HBD Formula	E (eV)	C1-C2 bond length (Å)	C2-C3 bond length (Å)
PM3	C ₃ H ₈ O ₃	-6.26	1.54708212	1.54304303
HF	C ₃ H ₈ O ₃	-9276.88	1.52494550	1.52000917
DFT	C ₃ H ₈ O ₃	-9381.60	1.53374476	1.52549611

The analysis of the results presented in Table 6 reveals that the different computational methods/levels assessed gave closely related values for their respective electronic energies, except for the PM3 method, which gave a higher value of -6.26 eV. This exceptional case identified for using PM3 could be due to the level of approximation inherent in semi-empirical calculations following existing reports in the literature ([51,54]). However, a probe into the distances between the atom length of the C1-C2 and C2-C3 bonds in the glycerol structure in Fig. 6 shows an insignificant difference in their bond distances for all the computational methods used. HF/DFT gave a bond distance of 1.53 Å for

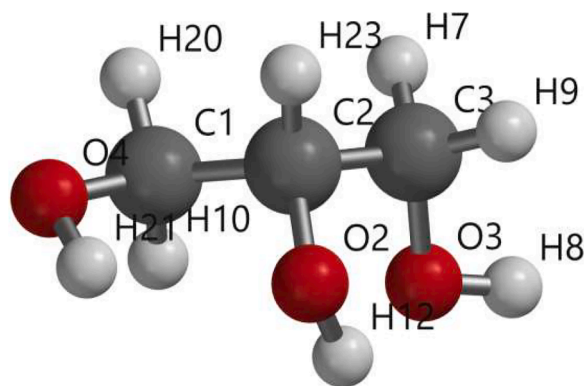


Fig. 6. The optimized geometry of glycerol (GCL) was obtained from DFT calculation.

the C1-C2 bond with a difference of 0.02 Å for PM3 calculation, and for the C2-C3 bond, there was a difference of 0.01 Å in their bond distance for PM3, HF and DFT methods respectively, which agreed with the literature report [25] for the rating of the methods' accuracy.

Analysis of the CHL-GLC deep eutectic solvent stability

In this section, we evaluate the stability of the DES formed by choline chloride (HBA) and glycerol (HBD) by probing the interatomic interaction of the atoms in the HBA (Cl⁻, H, N⁺, and O atom) with the HBD and also the HBA and HBD in their unbonded state during their structural formation. CHL and GLC, the HBA and HBD component of the DES, whose energetically stable formation mechanism explored in this report shows -21,472.65 and -9381.60 eV electronic energy, respectively.

Exploration of DES formation via the Cl-X

The analysis of the DES formation from CHL and GLC was evaluated for possible formation via the different interaction routes Cl-X (X = HO, OH, CH, HC, CH₂, H₂C) involving the Cl⁻ in CHL interacting with other components like O, H, and C present in the GLC (the HBD). In this analysis, several interactions evaluated include OH-Cl-HO (the interaction of Cl⁻ with two H-atoms of two different O-H functions in GLC), HO-Cl-OH (the interaction of Cl⁻ with two O-atoms from two O-H functions of GLC), CH-Cl-HO (that is, the interaction of Cl⁻ with two H-atoms from C-H and O-H functions of GLC), Cl-OH (the interaction of Cl⁻ with one O-atom in O-H functions of GLC), Cl-CH₂ (the interaction of Cl⁻ with one C-atoms from H-C-H function of GLC), Cl-HC (the interaction of Cl⁻ with one H-atom from C-H function of GLC), and physisorption without any constraint of a chemical bond in the model.

Findings from the study of the aforementioned interaction via the chloride ion have confirmed our study to have consistently restructured into a new geometrical structure, which favors the interaction of Cl⁻ with H-atom on O-H functions. The study reveals that some of the optimized structures for the DES formed via the various interaction routes investigated show a single, and some show multiple bonds with the different O-H functions present in GLC. Results shown in Table 7 are presented in the order of their interaction strength, where OH-Cl-HO was found

to have shown much stronger interaction compared to other interactions investigated in the study, as seen from its binding energy (BE) value. Fig. 7 presents the geometrical structure of the DES for the optimized or relaxed structure and the unrelaxed structure (that is, the structure before the geometrical optimization) of the best interaction (OH-Cl-HO) obtained for the investigation of Cl-X (X = HO, OH, CH, HC, CH₂, H₂C). Fig. 7 shows that the OH-Cl-HO bond lengths were shorter before optimization than after being optimized. The increase in the bond length of some interactions with bond lengths greater than 3 Å confirms them as weak interactions, contrary to other interactions that are stronger due to the shorter bond length obtained for the atom pairs, following the report of Pauling [49] that established the dependency of bond energy on its bond length. The strong interactions with bond lengths of around the range of 1.8 to 3 Å [13,30] were confirmed to be hydrogen bonding, while the weaker ones with bond lengths greater than 3 Å [40] were identified to be Van de Waal interaction, following the related reports [13,30,40] in the literature.

Further analysis of the bond length reports in Table 7 shows the strength of the OH-Cl-HO interaction, which was evident in its bond length obtained for its interaction with OH21-Cl1 and OH12-Cl1, which were found to be relatively low compared to other O-H like OH8 and OH1 that were longer. The bond length implies that there is a weak interaction of Cl with H8 and H1 compared to H21 and H12, which were stronger, in agreement with the literature [49] that established the dependency of bond energy on its bond length. A close range of shorter bond lengths was also found for the case of HO-Cl-OH and CH-Cl-HO, which also show a strong interaction with H21 and H12, unlike the case of other H-atoms (H8 and H1) in other O-H functions. Other interactions like Cl-OH, Cl-CH₂, and Cl-HC that were investigated were found to have shown much weaker interaction for all the H-atoms except for the one of the H-atom, which offers a better interaction due to its shorter bond length relatively found in other H-atoms of the alcohol function. So, in the evaluation of the Cl-X (X = HO, OH, CH, HC, CH₂, H₂C), the OH-Cl-HO shows the best interaction with a stronger binding energy of BE_{corr} = -0.49 eV and shorter bond length for Cl-atom with two other H-atoms of the alcohol (OH) functions in GLC, according to the literature [49].

Exploration of DES formation via the H-X

Further analysis of the interaction via the hydrogen atom of the HBA via the different interaction routes H-X (X = HO, OH, CH, CH₂, HC) as presented in Table 8, which favors the interaction of H with H-atom on O-H functions (OH-H-HO) with a binding energy of BE_{corr} = -0.79 eV when compared to other interaction routes and confirmed the stability and preference of interaction with H-atom on O-H functions as seen for using Cl⁻ as HBA. Fig. 8 presents the geometrical structure of the DES for the optimized or relaxed structure and the unrelaxed structure of the best and most stable interaction (OH-H-HO).

A probe into the optimized structure geometry suggests hydrogen bonding, as seen in the increase in bond length after optimization. Also, the strength of the OH-H-HO interaction in Table 8 was evident in the contribution of the interactions for OH12-Cl1, OH8-Cl1, and OH1-Cl1 in the shorter bond length (that is, stronger interaction) obtained when compared to OH21-Cl1 that shows longer bond length (that is, weaker interaction), in accordance with the literature [13,49]. The resulting

Table 7

Stability of the CHL-GLC DES via chloride (Cl⁻) ion as a point of interaction using DFT (Note that BE_{corr} is the BSSE corrected binding energies and BE_{uncorr} is the uncorrected binding energies).

Label	Interact	Name	BE _{corr} (eV)	BE _{uncorr} (eV)	Err (eV)	OH21-Cl1 (Å)	OH12-Cl1 (Å)	OH8-Cl1 (Å)	OH1-Cl1 (Å)
OH-Cl-HO	Cl	X6	-0.49	-0.52	0.03	2.27	2.26	6.21	6.02
HO-Cl-OH	Cl	X4	-0.46	-0.49	0.03	2.34	2.42	5.82	2.17
CH-Cl-HO	Cl	X3	-0.44	-0.48	0.04	2.22	2.36	5.61	5.89
Cl-OH	Cl	X5	-0.37	-0.48	0.11	2.08	5.70	6.57	4.11
Cl-CH ₂	Cl	X1	-0.35	-0.38	0.03	2.31	5.42	6.83	5.80
Cl-HC	Cl	X2	0.04	0.05	-0.01	5.87	5.52	5.08	2.41

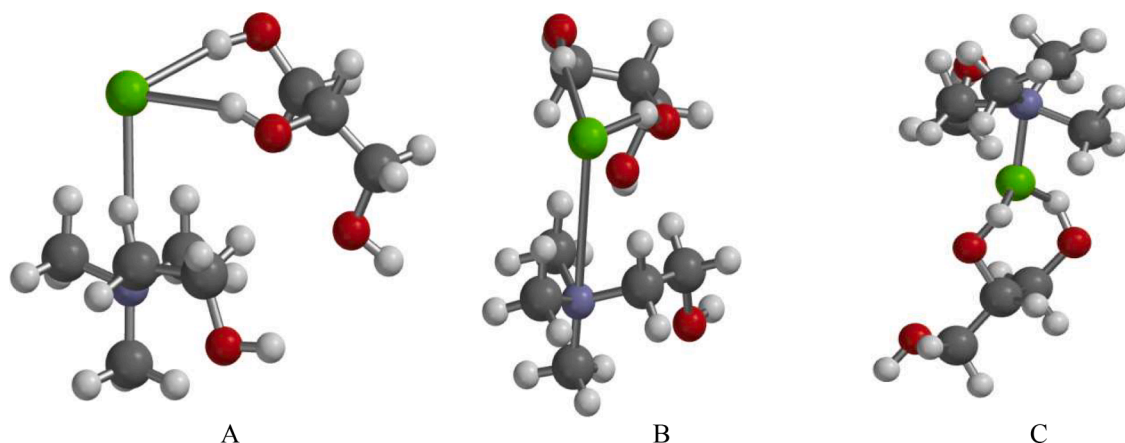


Fig. 7. The most stable geometric structure of the DES for the Cl-X interaction: (A) After geometrical optimization (one view), (B) After geometrical optimization (another view), and (C) Before geometrical optimization (one view).

Table 8

Stability of the CHL-GLC deep eutectic solvent via H as a point of interaction using DFT (Note that BEcorr is the BSSE corrected binding energies and BEncorr is the uncorrected binding energies).

Label	Interact	Name	BEcorr (eV)	BEncorr (eV)	Err (eV)	OH21-Cl1 (Å)	OH12-Cl1 (Å)	OH8-Cl1 (Å)	OH1-Cl1 (Å)
OH—H-HO	H	X6	−0.79	−0.95	0.16	5.03	2.16	2.21	2.10
CH-H-HO	H	X3	−0.62	−0.73	0.11	5.01	2.15	2.19	2.17
H-OH	H	X5	−0.59	−0.68	0.09	2.14	5.39	7.06	2.16
HO-H-OH	H	X4	−0.49	−0.56	0.07	2.25	2.28	5.43	2.23
H-CH2	H	X1	−0.44	−0.48	0.04	2.33	5.46	6.92	2.21
H-HC	H	X2	−0.34	−0.39	0.05	2.28	5.29	6.11	2.22

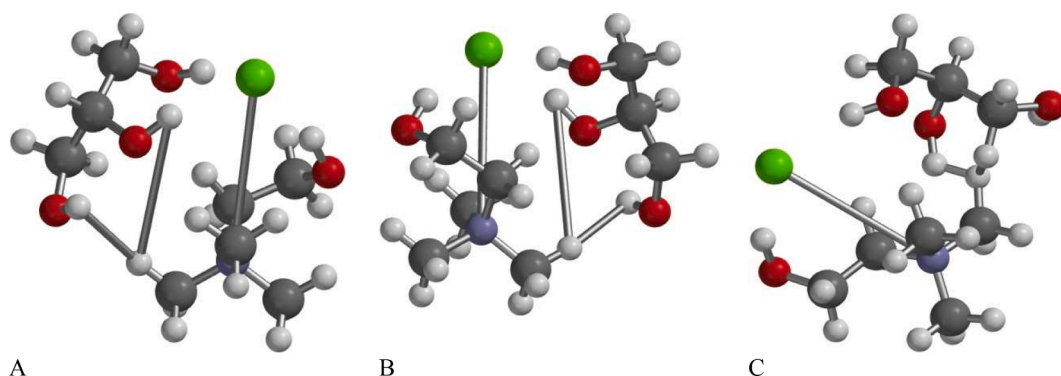


Fig. 8. The most stable geometric structure of the DES for the H-X interaction: (A) After geometrical optimization (one view), (B) After geometrical optimization (another view), and (C) Before geometrical optimization (one view).

geometrical structures obtained after the geometry optimization consistently favor the interaction of Cl-atom with H-atoms on the alcohol (OH) functions.

The short bond length implies that there is a strong interaction of the Cl with H12, H8 and H1, and the more this number of contributions from the H atom, the stronger the bond, which was evident in the lower

binding energy obtained for the best and most stable (OH—Cl-HO) for the H-X interaction (BEcorr = −0.79 eV) when compared to BEcorr = 0.49 eV which was obtained for the most stable in interaction for Cl-X (OH—Cl-HO) which has two strong interaction at H2 and H12.

Table 9

Stability of the CHL-GLC DES via nitrogen (N+) as a point of interaction using DFT (Note that BEcorr is the BSSE corrected binding energies and BEncorr is the uncorrected binding energies).

Label	Int	Name	BEcorr (eV)	BEncorr (eV)	Err (eV)	OH21-Cl1 (Å)	OH12-Cl1 (Å)	OH8-Cl1 (Å)	OH1-Cl1 (Å)
OH—N-HO	N	X6	−0.67	−0.78	0.11	2.26	2.15	5.28	2.23
N-CH2	N	X1	−0.50	−0.62	0.12	4.90	2.13	5.03	2.09
HO-N-OH	N	X4	−0.43	−0.54	0.11	4.36	7.33	9.55	2.07
N-HC	N	X2	−0.46	−0.52	0.06	2.09	6.56	7.37	4.06
N-HO	N	X5	−0.41	−0.49	0.08	2.15	2.45	5.47	2.32
CH-N-HO	N	X3	−0.10	−0.12	0.02	7.28	6.01	7.76	2.07

Exploration of DES formation via the N-X

Exploring N(+) as a point of interaction via the different interaction routes N-X ($X = \text{HO}, \text{OH}, \text{CH}, \text{CH}_2, \text{HC}$) as presented in Table 9, which are shown in order of their interaction strength. OH—N—HO was found to have a much stronger interaction than other interactions investigated with a binding energy of $\text{BE}_{\text{corr}} = -0.67 \text{ eV}$. The geometrical structure of the DES for the relaxed and the unrelaxed structure of best interaction at OH—N—HO shows that the relaxed structure had a shorter bond length when compared to the unrelaxed structure.

Further analysis of the bond length in Fig. 9 shows that the strength of the OH—N—HO interaction, which was evident in its bond length obtained for its interaction for OH21-Cl1, OH12-Cl1, and OH12-Cl1, which were found to be relatively low compared to OH8-Cl1, which was longer. This bond length implies a weak interaction of Cl with H8, while H21, H12, and H1 exhibited stronger interactions. It is important to note that the three (3) contributions from the H-atoms resulted to the stronger observed at H21, H12, and H1, which gave the strongest binding energy ($\text{BE}_{\text{corr}} = -0.67 \text{ eV}$) among other cases explored for H-X in the analysis. Other cases were seen to have consistently displayed a much longer bond length, which must have been resulting to their corresponding lower binding energies obtained from the analysis.

Our findings further agreed with the existing literature [13,49], indicating the magnitude of the bond distance between two atoms can give insight into the nature of bonding or interaction strength. This was also evident in the stronger binding energy obtained for the interaction via H-X ($\text{BE}_{\text{corr}} = -0.79 \text{ eV}$), which shows a much shorter bond length for most of its hydrogen bonding atoms compared to N-X ($\text{BE}_{\text{corr}} = -0.67 \text{ eV}$) shows a much longer bond length values with corresponding lower binding strength in Table 8 and 9, respectively.

Exploration of DES formation via the O-X

The results of exploring the use of O-atom as a point of interaction from the HBA through different interaction routes O-X ($X = \text{HO}, \text{OH}, \text{CH}, \text{CH}_2, \text{HC}$) of the HBD reveals that the interaction between O—OH was strongest when compared to other investigated interactions with a binding energy of $\text{BE}_{\text{corr}} = -0.63 \text{ eV}$ as presented in Table 10. Despite the best interaction being based on the oxygen, not the H in this case, however, the overall best interaction for the CHL-GLC deep eutectic solvent (DES) studied via Cl-X, H-X, N-X was strongest for H-X ($\text{BE}_{\text{corr}} = -0.79 \text{ eV}$) and through alcohol (HO) function.

Fig. 10 presents the geometrical structure of the CHL-GLC deep eutectic solvent (DES) for the optimized or relaxed structure and the unrelaxed structure of best interaction (O—OH) obtained for the investigation. A probe into the optimized structure, as shown in Fig. 10, shows that the strength of the O—OH interaction, which was evident in its bond length obtained for its interaction for OH12-Cl1 and OH1-Cl1, which were found to be relatively low compared to other O—H like

OH21 and OH8 that were longer (see Table 10). These bond lengths imply a weak interaction of Cl with H21 and H8 when compared to H12 and H1, which were stronger and are also shown in the smaller value of the bond length of 2.09 and 2.15 Å, respectively, in agreement with Pauling [49] report.

Thermodynamic analysis of the most stable DES formation pathways

The resulting outcome of the various pathways that could lead to the formation of the CHL:GLC deep eutectic solvent reported in Tables 7,8,9,10, where the most feasible path across each of the routes for Cl-X, H-X, N-X, and O-X were identified using their binding energies (BE) was employed in the thermodynamic analysis of the paths. Table 11 summarizes the best paths across each route studied and their corresponding outcome from the thermodynamic analysis carried out for the routes.

The results show that all the formation pathways interact with two H-atoms except for the H-X and N-X pathways that interact with three H-atoms, which showed higher binding strength for H-X and N-X, excluding the case of physisorption. The analysis further reveals that all the formation routes are spontaneous and exothermic due to the negative entropy and enthalpy changes obtained for all the routes, in accordance with the literature [6,23,37,55]. However, the trend of their degree of disorderliness and exothermicity was in the order of $\text{H-X} < \text{N-X} < \text{Phys} < \text{O-X} < \text{Cl-X}$ and $\text{H-X} > \text{Phys} > \text{N-X} > \text{O-X} > \text{Cl-X}$, confirming DES form via H-X to be the most exothermic and spontaneous formation pathway compared other pathways. With the use of the Gibbs formation energies, the exergonicity of the formation pathways was found to have followed the trend reported for the formation enthalpies in Table 11, in accordance with the literature [6,23,37,55]. The study confirms the H-X DES formation pathway, which involves the interaction of three H-atoms present in the alcohol functions of the HBD molecule (GLC), showing the strongest binding energy (or formation energy), most exothermic, spontaneous, and exergonic pathway leading to the formation of DES from CHL and GCL for any potential applications, in agreement with the literature [4,46,53,62], which shows that report the most negative binding energy to be the strongest interaction in a study.

This study's approach yielded more effective and straightforward findings compared to using maps for HOMO-LUMO energies and electrostatic potentials, which can be confusing and less precise. The results obtained from the electrostatic potential map (Fig. 12) in our quest to identify potential interaction points on the hydrogen bond donor (HBD), namely GLC, and the hydrogen bond acceptor (HBA), CHL, for the formation of a stable deep eutectic solvent (CHL:GLC) raised certain limitations.

Specifically, the points of interaction identified on the HBD (H on GLC) and HBA (Cl on CHL) molecules, as indicated by these maps, did not provide sufficient information to determine the number of hydrogen

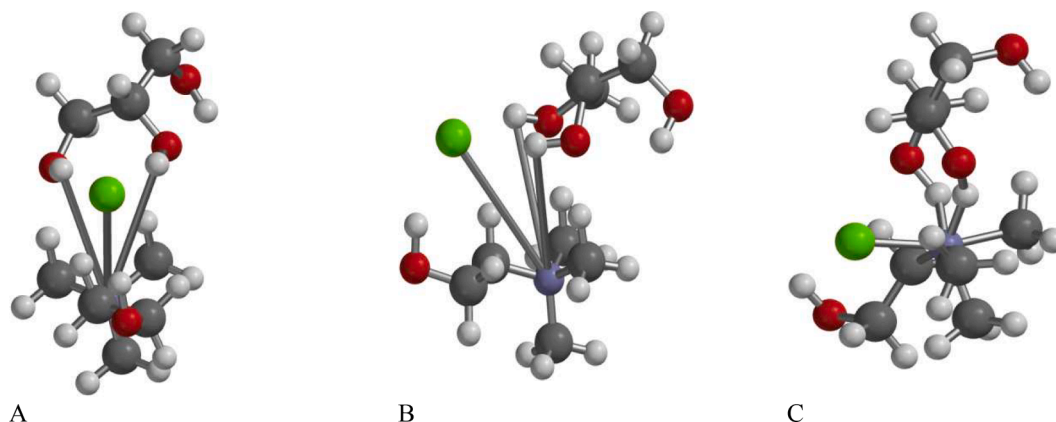


Fig. 9. The most stable geometric structure of the DES for the N-X interaction: (A) After geometrical optimization (one view), (B) After geometrical optimization (another view), and (C) Before geometrical optimization (one view).

Table 10

Stability of the CHL-GLY DES via oxygen(O) as a point of interaction using DFT (Note that BEcorr is the BSSE corrected binding energies and BEncorr is the uncorrected binding energies).

Label	Int	Name	BEcorr (eV)	BEncorr (eV)	Err (eV)	OH21-Cl1 (Å)	OH12-Cl1 (Å)	OH8-Cl1 (Å)	OH1-Cl1 (Å)
O-OH	O	X5	-0.63	-0.74	0.11	5.24	2.09	6.75	2.15
CH-O-HO	O	X3	-0.48	-0.57	0.09	2.48	3.95	6.30	2.01
O-HC	O	X2	-0.45	-0.51	0.06	2.12	5.60	7.08	2.25
HO-O-OH	O	X4	-0.39	-0.46	0.07	2.19	4.91	2.16	5.66
O-CH2	O	X1	-0.27	-0.38	0.11	4.61	6.21	3.81	2.10
OH-O-HO	O	X6	0.09	2.9E-3	0.09	6.36	7.70	10.12	6.95

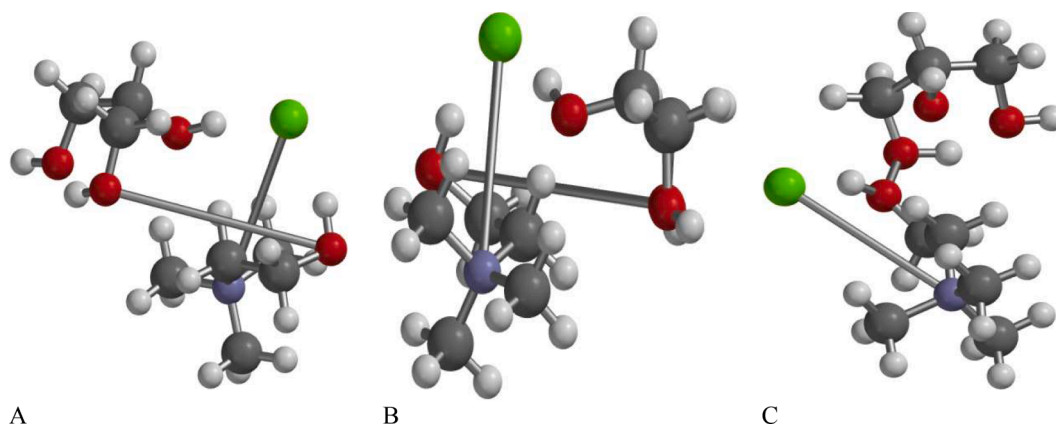


Fig. 10. The most stable geometric structure of the DES for the O-X interaction: (A) After geometrical optimization (one view), (B) After geometrical optimization (another view), and (C) Before geometrical optimization (one view).

Table 11

Thermodynamics analysis of possible interactions in the formation of CHL:GLC DES (Note that BEcorr is the BSSE corrected binding energies and BEncorr is the uncorrected binding energies).

Label	Interact	BEcorr (eV)	BEncorr (eV)	X in Cl-XO	dG° (eV)	dH° (eV)	dS° (meV/K)
OH-Cl-HO	Cl-X	-0.49	-0.52	H21, H12	0.11	-0.52	-2.11
OH-H-HO	H-X	-0.79	-0.95	H12, H8, H1	-0.25	-0.92	-2.25
Physio	N/A	-0.78	-0.89	H21, H12	-0.20	-0.85	-2.20
OH-N-HO	N-X	-0.67	-0.78	H21, H12, H1	-0.08	-0.74	-2.21
O-OH	O-X	-0.63	-0.74	H12, H1	-0.08	-0.73	-2.19

bonds in the HBD necessary for stable DES formation. This limitation observed in the use of maps underscores the importance of further research on deep eutectic solvents, considering the strategy employed in our study as a more promising avenue for exploration.

In general, the various DES formation pathways explored in our study diagrammatically shown in Fig. 11, which includes the Cl-X, H-X, N-X, and O-X, were confirmed to have all shown better interaction towards the formation of CHL:GLC via H-atoms of the glycerol (HBD) and Cl-atom of the choline chloride (HBA). The IR spectrum for the resulting CHL:GLC deep eutectic solvent obtained from the most energetically stable formation routes involving the interaction of Cl-atom with three other H-atoms of alcohol functions was also computed. The IR spectrum is presented in Fig. 11. The identified significant peaks in the computed IR include the 3460–3276 1/cm for the stretching of O—H towards Cl, 3090–2986 1/cm for the stretching of the C—H bond, and smaller peaks at 1482–591 1/cm, which shows a good agreement with an experiment 3291, 3027–2929, 1478–610 1/cm. The IR spectra have been able to validate the findings made for the computed energetically stable geometry and formation route for the CHL:GLC investigated, with the good agreement that our computations have shown with the ones reported by Rhamalia et al. [50] in an experiment.

Conclusions and recommendations

In this study, we successfully investigated the implications of using different approaches (using no bond, generic bond, or single bond) to model the electrovalent or ionic interactions present in a hydrogen-bond acceptor (HBA) molecule used in the synthesis of deep eutectic solvent (DES). The study identified a thermodynamically feasible pathway to an energetically stable DES using choline chloride (CHL) and urea (URE). Additionally, we thoroughly assessed the suitability of employing different levels of theoretical calculations like PM3, Hartree Fock (HF), and density functional theory (DFT).

Our study revealed that the use of no bond, generic, or single bond for modeling the ionic or electrovalent bonds in HBA molecules did not show any significant difference across the levels (PM3, HF, and DFT) of calculation deployed in our studies. Therefore, any of these approaches can be used in modeling with freedom; however, we recommend maintaining consistency by using only one approach in a particular study.

The findings from our study demonstrate that the H-X DES formation pathway, involving the interaction of three H-atoms present in the alcohol functions of the hydrogen bond donor (HBD) molecule (glycerol - GLC) with the chloride ion in the HBA molecule (choline chloride - CHL), has been identified as the thermodynamically feasible path for the CHL:GLC DES formation. The H-X pathway exhibited the most

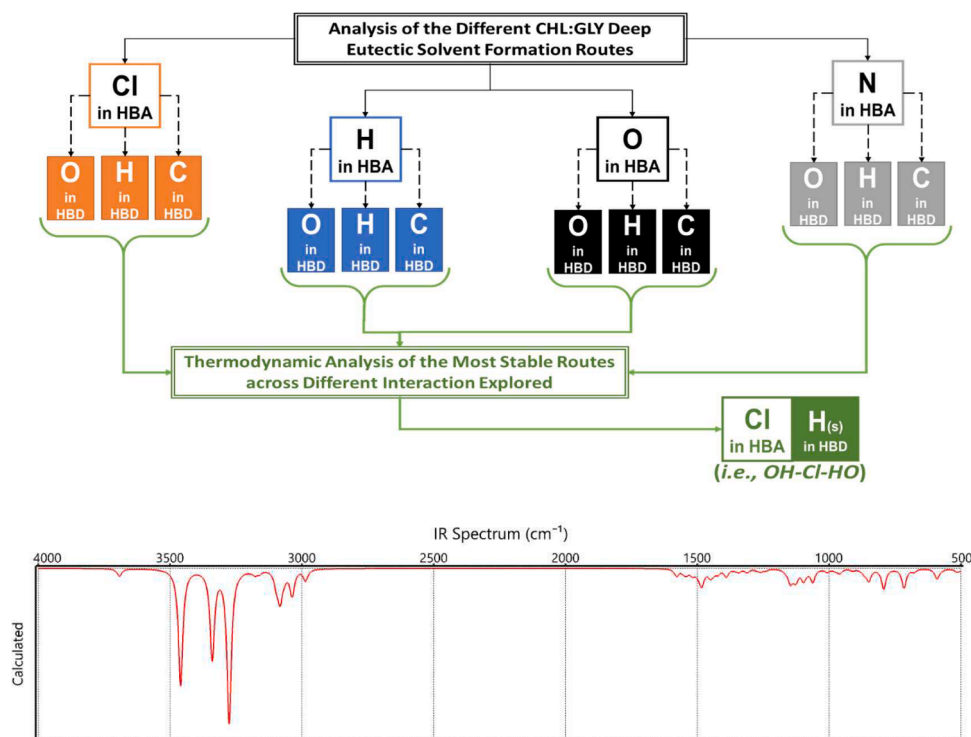


Fig. 11. A scheme showing the different DES formation routes explored the thermodynamically feasible one obtained from our study, including the IR spectra for the CHL:GLC deep eutectic solvent (DES) obtained for the most energetically stable formation routes.

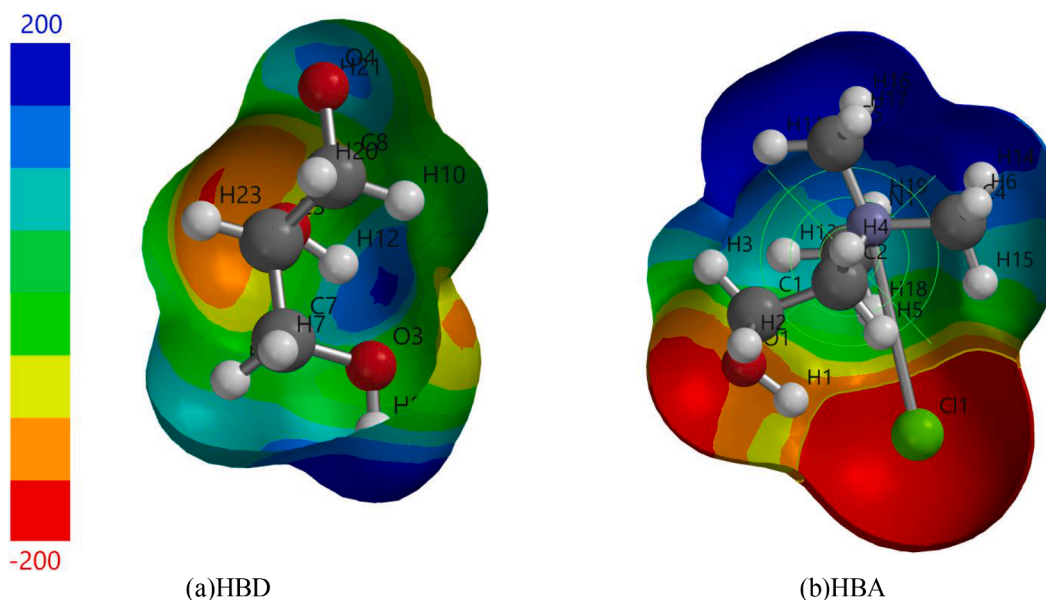


Fig. 12. A map showing the electrostatic potential interaction for glycerol (HBD) and choline chloride (HBA) for the most stable structures.

substantial binding energy (or formation energy), highest exothermicity, spontaneity, and exergonicity, leading to the formation of an energetically stable DES from CHL and GLC for any potential applications. Moreover, the deductions from different levels of calculation explored in our study support the validity of the H-X pathway consistently.

This study suggests that other levels of calculation can be employed in cases where it is expensive to afford DFT calculation in this kind of study. Our study findings recommend using our evaluation strategy for building different stable DES(s) that could be employed in the screening

of DES to best suit a defined application at both low and high levels of computational cost.

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CRediT authorship contribution statement

Maryann I. Uzochukwu: Formal analysis, Investigation, Software,

Resources, Visualization, Writing – original draft, Writing – review & editing. **Toyese Oyegoke**: Conceptualization, Formal analysis, Investigation, Software, Supervision, Methodology, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Raheem O. Momoh**: Project administration, Supervision, Validation. **Muhammed T. Isa**: Project administration, Supervision. **Suleiman M. Shuwa**: Project administration, Supervision. **Baba Y. Jibril**: Formal analysis, Investigation, Project administration, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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